Lego is a household name across the world, supported by a diverse toy line, hit movies, and a series of successful video games. In this project, we are going to explore a key development in the history of Lego: the introduction of licensed sets such as Star Wars, Super Heroes, and Harry Potter.

The introduction of its first licensed series, Star Wars, was a hit that sparked a series of collaborations with more themed sets. The partnerships team has asked you to perform an analysis of this success, and before diving into the analysis, they have suggested reading the descriptions of the two datasets to use, reported below.

## The Data

You have been provided with two datasets to use. A summary and preview are provided below.

### lego_sets.csv

| Column | Description |
|---|---|
| `"set_num"` | A code that is unique to each set in the dataset. This column is critical, and a missing value indicates the set is a duplicate or invalid! |
| `"name"` | The name of the set. |
| `"year"` | The date the set was released. |
| `"num_parts"` | The number of parts contained in the set. This column is not central to our analyses, so missing values are acceptable. |
| `"theme_name"` | The name of the sub-theme of the set. |
| `"parent_theme"` | The name of the parent theme the set belongs to. Matches the name column of the parent_themes csv file. |

### parent_themes.csv

| Column | Description |
|---|---|
| `"id"` | A code that is unique to every theme. |
| `"name"` | The name of the parent theme. |
| `"is_licensed"` | A Boolean column specifying whether the theme is a licensed theme. |

```
# Import pandas, read and inspect the datasets
import pandas as pd

lego_sets = pd.read_csv('data/lego_sets.csv')
lego_sets.head()
```

| index | set_num | name | year | num_parts |
|---|---|---|---|---|
| 0 | 00-1 | Weetabix Castle | 1970 | |
| 1 | 0011-2 | Town Mini-Figures | 1978 | |
| 2 | 0011-3 | Castle 2 for 1 Bonus Offer | 1987 | |
| 3 | 0012-1 | Space Mini-Figures | 1979 | |
| 4 | 0013-1 | Space Mini-Figures | 1979 | |

Rows: 5 Expand

```
parent_themes = pd.read_csv('data/parent_themes.csv')
parent_themes.head()
```

| index | id | name | is |
|---|---|---|---|
| 0 | 1 | Technic | F |
| 1 | 22 | Creator | F |
| 2 | 50 | Town | F |
| 3 | 112 | Racers | F |
| 4 | 126 | Space | F |

Rows: 5 Expand

```python
# Import pandas and read in the DataFrame, and inspect it
import pandas as pd
lego_sets = pd.read_csv('data/lego_sets.csv')
lego_sets.head()

# Drop relevant missing rows
lego_sets_clean = lego_sets.dropna(subset=['set_num', 'name', 'theme_name'])
lego_sets_clean.head()

# Get list of licensed sets
parent_themes = pd.read_csv('data/parent_themes.csv')
licensed_themes = parent_themes[parent_themes['is_licensed']]['name']
licensed_themes.head()

# Subset for licensed sets
licensed = lego_sets_clean['parent_theme'].isin(licensed_themes)
licensed_sets = lego_sets_clean[licensed]
licensed_sets.head()

# Calculate the percentage of licensed sets that are Star Wars themed
all_sets = len(licensed_sets)
star_wars_sets = len(licensed_sets[licensed_sets['parent_theme'] == 'Star Wars'])
ratio = star_wars_sets / all_sets
the_force = int(ratio * 100)
print(f'The percentage of licensed sets that are Star Wars themed is {the_force}%.')

# Create a pivot table of sets released by theme per year
licensed_pivot = licensed_sets.pivot_table(index='year', columns='parent_theme', values='set_num', aggfunc='count')

# Find the year when the most Star Wars sets were released
licensed_pivot.sort_values(by="Star Wars", ascending=False)["Star Wars"]
new_era = 2016
print(f'The year when the most Star Wars sets were released was {new_era}.')
```

```
The percentage of licensed sets that are Star Wars themed is 51%.
The year when the most Star Wars sets were released was 2016.
```